

**Software Engineering Department
Braude College of Engineering
Capstone Project - Phase B (61999)**

**Identifying patterns of ASD
behavior using Eye Tracking Technology**

Project Code: 26-1-D-1



By:

**Guy Zamir
Shani Fahima**

Advisor:

**Dr. Anat Dahan
Dr. Julia Sheidin**

Link to GitHub:

<https://github.com/GuyZamir12/asd-eye-tracking.git>

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1. Introduction

This Phase B project book summarizes the research project developed for identifying measurable differences between Autism Spectrum Disorder (ASD) and typically developing (TD) children using eye-tracking data. ASD is characterized by social-communication differences and restricted/repetitive behavior patterns, and eye-tracking has been proposed as an objective, non-invasive tool for quantifying visual-attention behavior [Hirota et al., 2023; Hodis et al., 2025; Hessels et al., 2024]. The project focuses on naturalistic visual behavior and translates raw Tobii eye-tracking outputs into interpretable computational features.

The implemented Python project processes eye-tracking CSV files, removes invalid samples and blinks, extracts participant-level gaze and pupil features, performs an ablation study, trains Random Forest classifiers using 5-fold stratified cross-validation, and produces classification reports, confusion matrices, and feature-pattern summaries.

The experimental results show that the combined feature set achieved the highest cross-validation accuracy of 74.09%. Pupil dynamics alone achieved 71.97% accuracy and the highest ASD recall, while exploration and exploitation features alone achieved 70.30% accuracy. The feature analysis indicates that ASD participants in the processed dataset generally showed lower fixation duration and lower pupil-diameter measures, but higher gaze dispersion, longer saccade length, and more AOI changes than TD participants.

The project is not intended to replace clinical diagnosis. It is a research-oriented decision-support framework that demonstrates how interpretable eye-tracking measures can be transformed into computational evidence for future ASD screening and behavioral-analysis tools. This wording follows the Phase A research motivation, which emphasized both the diagnostic potential of eye tracking and the need for careful interpretation and standardization before clinical use [Setien-Ramos et al., 2022; Eretova et al., 2024; Billeci et al., 2024].

2. Problem Review and Background

2.1 Research problem

ASD is a neurodevelopmental condition characterized by persistent difficulties in social communication and interaction, as well as restricted and repetitive patterns of behavior [Hirota et al., 2023]. ASD typically appears in early childhood, and early identification is important because timely, personalized intervention can improve long-term developmental outcomes [Hodis et al., 2025; Hirota et al., 2023]. Diagnosis is commonly based on clinical and behavioral observation, but this process can be time-consuming and dependent on expert interpretation. Therefore, there is a need for objective, quantifiable behavioral indicators that can support, but not replace, clinical assessment.

Eye tracking provides a non-invasive way to record visual attention and cognitive processing. It captures where a child looks, for how long, how gaze shifts between areas, and how physiological indicators such as pupil diameter change over time. Effective eye-tracking research requires selecting gaze metrics that align with the theoretical research question, ensuring that the extracted features reflect meaningful cognitive and behavioral processes [Hessels et al., 2024].

2.2 Gap addressed by the project

Previous eye-tracking studies have shown that individuals with ASD may exhibit atypical visual attention patterns, including reduced attention to social stimuli and circumscribed attention to specific non-social elements [Sasson et al., 2008; Vacas et al., 2021; Congiu et al., 2024]. Meta-analytic findings also indicate that gaze metrics can help distinguish ASD from non-ASD groups [Setien-Ramos et al., 2022]. However, many studies rely on static metrics such as total fixation duration on predefined Areas of Interest (AOIs), and the field still faces variability in experimental protocols and a lack of standardized gaze metrics [Eretova et al., 2024]. Our project addresses this gap by combining fixation, saccade, AOI, gaze-dispersion, and pupil-dynamics features into a structured analysis pipeline.

2.3 Research question

The main research question is: Can ASD related behavioral patterns be identified from eye movements and pupil dynamics during naturalistic visual exploration?

2.4 Project objective

The objective is to develop a data-driven research framework that compares ASD and TD participants using eye-tracking features, evaluates whether these features support classification, and provides interpretable outputs for researchers. The system is not presented as a clinical diagnostic device. It is a research prototype intended to support pattern discovery and future model development.

3. General Description

3.1 System purpose

The system is a data-analysis project for eye-tracking research. It receives CSV files exported from Tobii Pro Lab, processes the recordings, calculates participant-level features, compares ASD and TD groups, and evaluates classification performance using machine learning. The design follows the research direction described in Phase A: combining eye-tracking technology with ML/DL-style analysis to identify interpretable patterns of exploration and exploitation in ASD behavior [Gameiro et al., 2017; Eretova et al., 2024].

3.2 Intended users

- Researchers studying ASD-related gaze behavior.
- Clinical collaborators interested in objective behavioral indicators.
- Future students or maintainers who may extend the research pipeline.

3.3 Current scope

The current version is a research project rather than a clinical product. It supports offline analysis of existing CSV recordings, participant-level feature extraction, model evaluation, and result visualization. It does not provide real-time diagnosis, a clinical user interface, or medical recommendations.

4. Research Solution and System Structure

4.1 High-level solution

The solution is implemented as a Python analysis pipeline.

The pipeline follows six main stages: data loading, preprocessing, feature extraction, feature grouping, model training/evaluation, and interpretation of group-level patterns (Figure 1). This structure was selected because eye-tracking data must first be cleaned and transformed into theoretically meaningful metrics before it can support valid computational modeling [Hessels et al., 2024; Grabinger et al., 2024].



Research prototype pipeline implemented in Python

Figure 1. Research pipeline for eye-tracking analysis and ASD/TD classification.

4.2 System structure

The software is organized around data-processing and analysis modules rather than a fully deployed application.

The main components are the CSV loader, preprocessing logic, feature extraction logic, machine-learning pipeline, evaluation module, and visualization/reporting outputs (Figure 2).

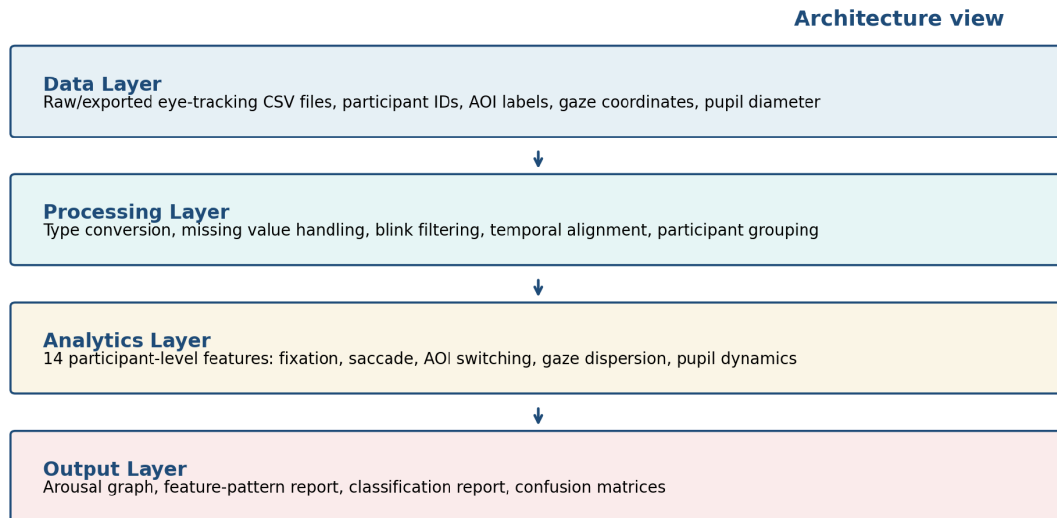


Figure 2. High-level structure of the Phase B research prototype.

4.3 Main algorithmic steps

1. Load all eye-tracking CSV files from the configured folder.
2. Validate that required columns exist and convert numerical columns to numeric types.
3. Remove invalid rows, missing timestamps, missing gaze values where needed, and blink samples.
4. Assign participants to ASD or TD groups according to dataset participant IDs.
5. Group consecutive eye-movement categories into behavioral blocks.

6. Extract exploration/exploitation features from fixations, saccades, AOIs, and gaze dispersion.
7. Extract pupil-dynamics features from pupil diameter values and temporal changes.
8. Train and evaluate Random Forest classifiers using stratified 5-fold cross-validation.
9. Run an ablation study to compare feature groups.
10. Generate classification reports, confusion matrices, and feature-pattern summaries.

4.4 Feature groups implemented

Table 1 summarizes the 14 participant-level features implemented in the system and organizes them into three analysis groups. The exploration and exploitation group captures visual-attention behavior through fixation, saccade, AOI-transition, and gaze-dispersion measures. This grouping is based on the distinction between broad spatial sampling and deeper processing of a currently fixated region [Gameiro et al., 2017]. In addition, the use of AOI changes and gaze-transition measures is consistent with recent research emphasizing dynamic visual-attention analysis rather than relying only on static fixation summaries [Ellis et al., 2021; Wang et al., 2024]. The pupil-dynamics group represents physiological and cognitive changes using pupil size and variability measures. Finally, the combined feature group includes all 14 features and is used to examine whether integrating gaze behavior with pupil-based information improves ASD and TD classification.

Feature Group	Implemented Features	Research Meaning
Exploration and exploitation	Average fixation duration, maximum fixation duration, total AOI time, saccade frequency, average saccade length, AOI changes, gaze dispersion.	Represents focus depth, scanning breadth, and spatial exploration.
Pupil dynamics	Average pupil, maximum pupil, standard deviation, coefficient of variation, pupil range, pupil velocity, fixation pupil.	Represents physiological/cognitive load and pupil-based variability.
Combined features	All 14 features together.	Tests whether combining gaze behavior and pupil dynamics improves classification.

4.5 Functional requirements supported by the current prototype

Table 2 presents the main functional requirements:

Requirement	Current Status
FR1 External Data Ingestion	Supported. The system imports exported CSV files from a configured folder.
FR2 Data Cleaning and Filtering	Supported. The prototype removes missing data and blink samples.

FR3 AOI Mapping	Partially supported. The prototype uses the AOI Name Right column when it exists; full automatic raw-coordinate-to-AOI mapping remains a future enhancement.
FR4 Time-Series Generation	Partially supported. The arousal module builds sliding-window time series; a full transition-sequence dataset for LSTM is future work.
FR5 Exploration and Exploitation Feature Extraction	Supported. Participant-level exploration and exploitation features are calculated.
FR6 ML/DL Classification	Supported for ML baseline. Random Forest classification is implemented; LSTM/hybrid DL remains planned.
FR7 Reporting and Interpretation	Supported. The code prints reports and generates graphs, confusion matrices, and feature pattern summaries.

4.6 Non-functional requirements

Table 3 presents the main non-functional requirements:

Quality Attribute	Project Response
Accuracy and performance	Initial cross-validated accuracy reaches approximately 75% with exploration/exploitation features. Further validation is required.
Scalability	Current script can process multiple CSV files. For larger datasets, modularization and vectorized processing should be strengthened.
Data security and privacy	The software should be operated only on de-identified participant files and stored in approved research locations.
Maintainability	The current script is readable and commented; future work should split it into modules and configuration files.

5. Research and Development Process

5.1 Experimental and Data Context

The original Phase A research plan was based on a naturalistic playroom experiment conducted in collaboration with researchers from the University of Haifa. In the planned experiment, children were expected to play freely in a dedicated playroom while wearing Tobii eye-tracking glasses. The recordings were intended to include gaze coordinates, fixation and saccade categories, pupil diameter values, and AOI-related fields defined

according to the physical areas and objects in the room. This setup was designed to support the project goal of studying spontaneous visual exploration and naturalistic behavior in children with ASD compared to typically developing children. However, during Phase B, the University of Haifa experiment was still in a very early stage, and the full experimental dataset was not yet available for computational analysis. Therefore, to continue developing and validating the analytical framework, we used a backup eye-tracking dataset [Cilia, et al., 2022]. This dataset includes recordings from participants who viewed visual stimuli, such as images or videos, on a computer screen while their eye movements were recorded. The project was therefore implemented and evaluated using the available backup dataset. After preprocessing and filtering invalid or incomplete records, the final dataset included 57 valid participants: 30 TD participants and 27 participants diagnosed with ASD. Although this dataset is less naturalistic than the planned playroom experiment, it still provides relevant eye-tracking signals, including fixation behavior, saccadic activity, gaze dispersion, AOI transitions, and pupil dynamics. These signals allowed us to develop and test the main analysis pipeline and to examine whether measurable eye-tracking patterns can distinguish between ASD and TD groups.

5.2 Development workflow

The development workflow followed a structured sequence, beginning with data inspection and preprocessing, continuing with feature extraction and model evaluation, and ending with the interpretation of the results and formulation of the project conclusions:

1. Inspect the Tobii CSV structure and identify required columns.
2. Implement preprocessing for numeric conversion, missing data, and blink filtering.
3. Implement feature extraction for fixation, saccade, AOI, gaze dispersion, and pupil values.
4. Construct three feature sets for ablation testing.
5. Evaluate models using Random Forest, MinMax scaling, grid search, and 5-fold stratified cross-validation.
6. Analyze outputs and make project conclusions.

5.3 Research logic

The research logic is based on the assumption that ASD and TD groups may differ not only in where they look, but also in how they explore a visual environment. Therefore, the pipeline combines focus-related, scanning-related, AOI transition-related, and pupil-related measures.

This is consistent with research showing the importance of exploration/exploitation behavior, scanpath variability, and dynamic gaze patterns in ASD-related visual attention [Gameiro et al., 2017; Falck-Ytter et al., 2013; Ellis et al., 2021]. The ablation study helps determine which feature family contributes most to classification performance.

6. Tools and Customer Interface

6.1 Tools used

Tool	Purpose in Project
Python	Main implementation language for the research project.
Google Colab	Development and execution environment.
pandas	CSV loading, data cleaning, grouping, and tabular processing.
NumPy	Numerical calculations and feature arrays.
scikit-learn	MinMaxScaler, RandomForestClassifier, GridSearchCV, StratifiedKFold, classification reports, and confusion matrices.
matplotlib / seaborn	Visualization of confusion matrices and group-level feature/instability patterns.
Tobii Pro Lab	Source of exported eye-tracking CSV data and AOI-related columns.
GitHub	Repository for code and project deliverables.

7. Testing, Evaluation, and Results

7.1 Testing strategy

The project was tested at both data-processing and model-evaluation levels. Data-processing checks verified that required columns exist, numerical fields are converted correctly, participants are assigned to the correct group, and missing or blink samples do not distort the feature calculations. Model evaluation used 5-fold stratified cross-validation to preserve the proportions of ASD and TD across folds.

The evaluation uses participant-level classification rather than sample-level classification, which reduces the risk of data leakage and better represents the research goal: distinguishing participants based on summarized eye-tracking behavior.

7.2 Ablation study results

An ablation study was performed to evaluate the contribution of each feature group. The model was trained separately on exploration/exploitation features, pupil-dynamics features, and the full combined set of 14 features. This experiment was implemented in the Python project using scikit-learn Random Forest models, GridSearchCV, and 5-fold stratified cross-validation.

Feature Set	CV Accuracy	TD P/R/F1	ASD P/R/F1	Confusion Matrix Summary
Exploration & Exploitation Only	70.30%	0.68 / 0.83 / 0.75	0.75 / 0.56 / 0.64	TD: 25 correct, 5 errors; ASD: 15 correct, 12 errors
Pupil Dynamics Only	71.97%	0.75 / 0.70 / 0.72	0.69 / 0.74 / 0.71	TD: 21 correct, 9 errors; ASD: 20 correct, 7 errors
Combined Features (All 14)	74.09%	0.71 / 0.83 / 0.77	0.77 / 0.63 / 0.69	TD: 25 correct, 5 errors; ASD: 17 correct, 10 errors

The combined feature set achieved the best overall accuracy at 74.09%. Pupil dynamics alone achieved slightly higher performance than exploration/exploitation alone and showed the strongest ASD recall (0.74), indicating it correctly detected more ASD participants. However, the combined set achieved the best overall balance and the highest TD recall (Figure 3).

```

... Extracting Exploration, Exploitation, and Pupil features (Participant Level)...

Dataset ready! Total Participants: 57

=====
=== Feature Group Testing (Ablation Study) ===
=====

[ Evaluating: 1. Exploration & Exploitation Only ]
Best Params Found: {'clf__max_depth': 5, 'clf__min_samples_leaf': 1, 'clf__min_samples_split': 5, 'clf__n_estimators': 50}
Average CV Accuracy: 70.30%

Classification Report:
      precision    recall  f1-score   support

      TD         0.68      0.83      0.75        30
      ASD         0.75      0.56      0.64        27

   accuracy          0.70        57
  macro avg         0.71      0.69      0.69        57
 weighted avg         0.71      0.70      0.70        57


[ Evaluating: 2. Pupil Dynamics Only ]
Best Params Found: {'clf__max_depth': None, 'clf__min_samples_leaf': 1, 'clf__min_samples_split': 2, 'clf__n_estimators': 100}
Average CV Accuracy: 71.97%

Classification Report:
      precision    recall  f1-score   support

      TD         0.75      0.70      0.72        30
      ASD         0.69      0.74      0.71        27

   accuracy          0.72        57
  macro avg         0.72      0.72      0.72        57
 weighted avg         0.72      0.72      0.72        57


[ Evaluating: 3. Combined Features (All 14) ]
Best Params Found: {'clf__max_depth': None, 'clf__min_samples_leaf': 2, 'clf__min_samples_split': 2, 'clf__n_estimators': 100}
Average CV Accuracy: 74.09%

Classification Report:
      precision    recall  f1-score   support

      TD         0.71      0.83      0.77        30
      ASD         0.77      0.63      0.69        27

   accuracy          0.74        57
  macro avg         0.74      0.73      0.73        57
 weighted avg         0.74      0.74      0.73        57

```

Figure 3. Classification reports for the three feature groups.

7.3 Confusion matrix interpretation

The confusion matrices show that the combined model correctly classified 25 out of 30 TD participants and 17 out of 27 ASD participants. The pupil-only model detected 20 out of 27 ASD participants, but it misclassified more TD participants as ASD. This suggests that pupil features are informative for ASD detection, whereas the combined feature set yields higher overall accuracy.

However, the small sample size is a significant limitation that likely restricted the model's ability to learn more robust patterns, which explains the misclassifications and the lower overall diagnostic performance (Figure 4).

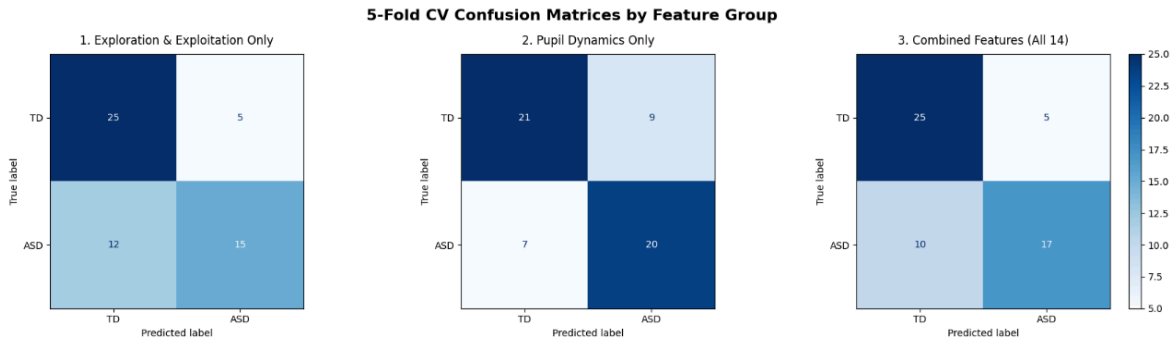


Figure 4. 5-fold cross-validation confusion matrices by feature group.

7.4 Feature-pattern analysis

The participant-level feature comparison provides an interpretable explanation of the model outputs. The table below presents the ASD and TD averages and the difference between them.

Feature	ASD Avg.	TD Avg.	ASD - TD
Avg Fixation Duration	361.6916	466.7981	-105.1066
Max Fixation Duration	2353.2244	2584.5713	-231.3469
Total AOI Time	1755.6667	2886.8667	-1131.2000
Saccade Frequency	1.3892	1.5693	-0.1801
Avg Saccade Length	311.1695	262.5534	+48.6161
AOI Changes	42.5185	38.4000	+4.1185
Gaze Dispersion	329.3335	265.1608	+64.1727
Avg Pupil	1.8910	2.9122	-1.0213
Max Pupil	2.3837	3.4592	-1.0756
Std Pupil	0.1848	0.3273	-0.1425
Pupil CV	0.0465	0.0813	-0.0347
Pupil Range	1.5319	2.5227	-0.9908
Pupil Velocity	1.9108	4.7807	-2.8700
Fixation Pupil	1.9002	2.9440	-1.0438

The feature-pattern results can be summarized as follows:

- **Exploitation/focus depth:**
ASD participants generally showed lower average and maximum fixation durations and substantially lower total AOI time.
- **Exploration/scanning breadth:**
ASD participants generally showed lower saccade frequency, but higher average saccade length, more AOI changes, and higher gaze dispersion.
- **Pupil dynamics:**
ASD participants generally showed lower pupil-diameter and pupil-variability measures than TD participants .

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=== Feature Patterns (ASD vs. TD Averages) ===
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[ ALL FEATURES COMBINED ]
Diagnosis      ASD      TD      Difference (ASD - TD)
Avg Fixation Duration 361.6916 466.7981 -105.1066
Max Fixation Duration 2353.2244 2584.5713 -231.3469
Total AOI Time      1755.6667 2886.8667 -1131.2000
Saccade Frequency    1.3892    1.5693    -0.1801
Avg Saccade Length    311.1695 262.5534  48.6161
AOI Changes          42.5185   38.4000   4.1185
Gaze Dispersion      329.3335 265.1608  64.1727
Avg Pupil            1.8910    2.9122    -1.0213
Max Pupil            2.3837    3.4592    -1.0756
Std Pupil            0.1848    0.3273    -0.1425
Pupil CV             0.0465    0.0813    -0.0347
Pupil Range          1.5319    2.5227    -0.9908
Pupil Velocity        1.9108    4.7807    -2.8700
Fixation Pupil        1.9002    2.9440    -1.0438

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=== Plain-English Pattern Breakdown (Categorized) ===
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[ EXPLOITATION (Focus & Depth) ]
▼ Avg Fixation Duration: ASD generally had LOWER values (-22.5% compared to TD).
▼ Max Fixation Duration: ASD generally had LOWER values (-9.0% compared to TD).
▼ Total AOI Time: ASD generally had LOWER values (-39.2% compared to TD).

[ EXPLORATION (Scanning & Breadth) ]
▼ Saccade Frequency: ASD generally had LOWER values (-11.5% compared to TD).
▲ Avg Saccade Length: ASD generally had HIGHER values (+18.5% compared to TD).
▲ AOI Changes: ASD generally had HIGHER values (+10.7% compared to TD).
▲ Gaze Dispersion: ASD generally had HIGHER values (+24.2% compared to TD).

[ PUPIL DYNAMICS (Cognitive Load & Arousal) ]
▼ Avg Pupil: ASD generally had LOWER values (-35.1% compared to TD).
▼ Max Pupil: ASD generally had LOWER values (-31.1% compared to TD).
▼ Std Pupil: ASD generally had LOWER values (-43.5% compared to TD).
▼ Pupil CV: ASD generally had LOWER values (-42.7% compared to TD).
▼ Pupil Range: ASD generally had LOWER values (-39.3% compared to TD).
▼ Pupil Velocity: ASD generally had LOWER values (-60.0% compared to TD).
▼ Fixation Pupil: ASD generally had LOWER values (-35.5% compared to TD).

```

Figure 5. Feature-pattern summary printed by the Python project.

The graph was generated using SHAP values calculated for the combined Random Forest model trained on all 14 features. For each participant, SHAP estimates how strongly each feature contributed to the model's prediction, and the absolute SHAP values were then averaged across all participants to obtain a global importance score. The features were ranked from the highest to the lowest average impact. The results show that pupil standard deviation was the most influential feature, followed by total AOI time, saccade frequency, average fixation duration, and gaze dispersion. This indicates that both pupil dynamics and gaze-behavior measures contributed meaningfully to the classification. In contrast, AOI changes and maximum fixation duration had relatively little influence on the model (Figure 6).

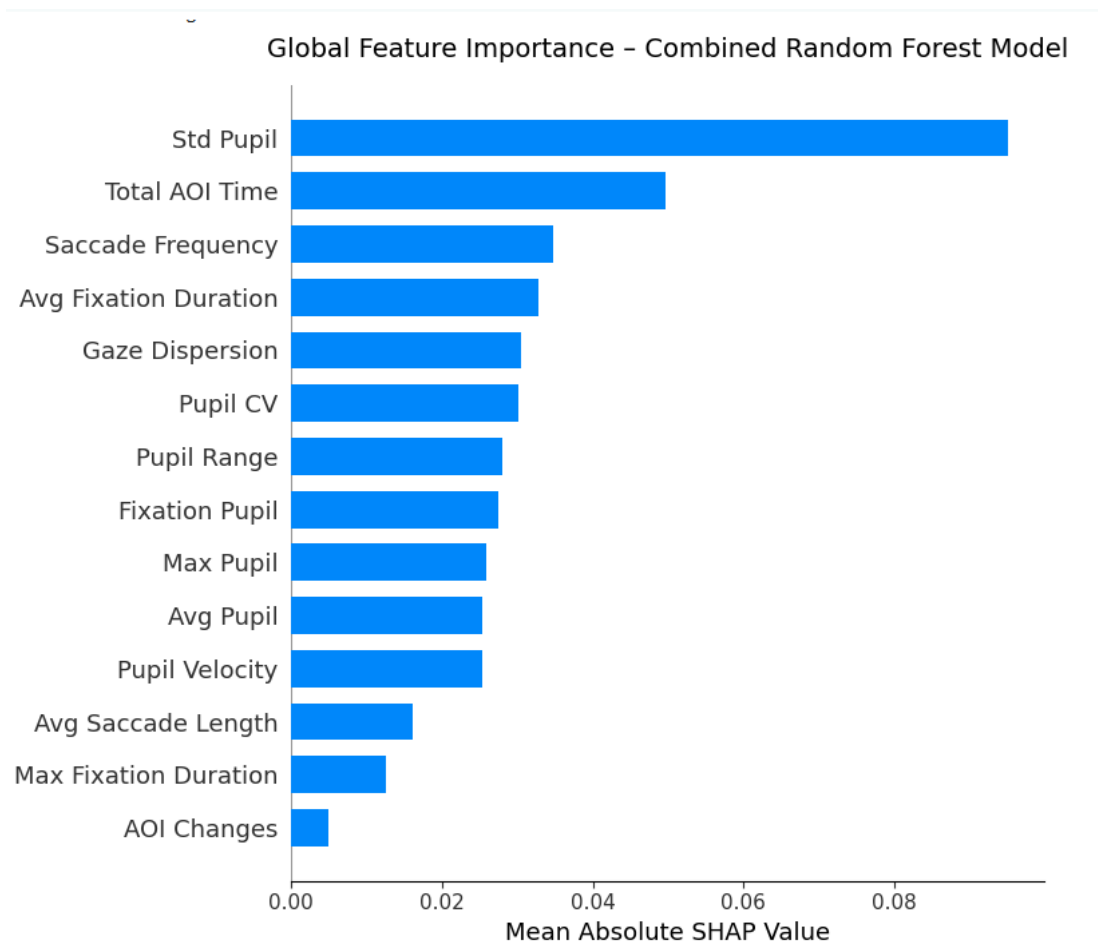


Figure 6. Global feature importance.

7.5 Pupil Instability and Arousal Analysis

As part of the pupil dynamics analysis, we examined short-term changes in pupil diameter as an indicator of physiological arousal and instability over time. The eye-tracking recordings were divided into overlapping 1-second windows, with a step size of 250 milliseconds between consecutive windows. For each valid window, the average pupil diameter was calculated, and the instability score was defined as the absolute change between the current window's average pupil diameter and that of the previous valid window. Calculated as follows:

$$\text{instability} = \text{abs}(\text{curr_mean} - \text{prev_mean})$$

In this calculation, curr_mean represents the average pupil diameter within the current time window, and prev_mean represents the average pupil diameter from the immediately preceding valid window.

Blink samples, missing values, and invalid records were removed before the calculation in order to reduce noise and avoid artificial fluctuations. The results showed that the ASD group had a higher average pupil instability score than the TD group. Specifically, the average pupil instability was 0.06349 mm in the ASD group, compared with 0.04915 mm in

the TD group. The difference between the groups was 0.01435 mm (Figure 7). This suggests that, in this dataset, participants in the ASD group demonstrated stronger short-term fluctuations in pupil diameter, which may reflect higher physiological arousal or less stable pupil dynamics during the visual task. Although ASD participants showed lower overall variability across the recording (standard deviation and coefficient of variation), they exhibited greater short-term temporal fluctuations, reflected by the pupil instability measure. The group-level graph further illustrates this difference over time. Although both groups show variability throughout the task, the ASD curve includes more frequent and higher instability peaks. These findings support the inclusion of pupil-based features in the analysis pipeline, since pupil dynamics may provide additional information beyond gaze position, fixation duration, and saccadic behavior.

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... === Arousal (Pupil Instability) Numerical Results ===
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ASD Average Pupil Instability (Change per 250ms): 0.06349 mm
TD Average Pupil Instability (Change per 250ms): 0.04915 mm

=> Result: The ASD group shows HIGHER fluctuation frequency (higher arousal).
    Difference: 0.01435 mm
=====

```

Figure 7. Arousal (pupil instability) numerical results.

The graph presents changes in average pupil instability over time for the ASD and TD groups, calculated using overlapping 1-second windows with 250-millisecond steps. The horizontal axis represents the relative task time in seconds from the beginning of each recording, while the vertical axis represents the instability score, defined as the absolute change in average pupil diameter between two consecutive windows, measured in millimeters. The red line represents the ASD group, and the blue line represents the TD group. Both groups show fluctuations throughout the task. However, the ASD group generally exhibits more frequent and higher instability peaks than the TD group. This pattern may indicate less stable pupil dynamics and stronger short-term changes in physiological arousal among ASD participants in this dataset (Figure 8).

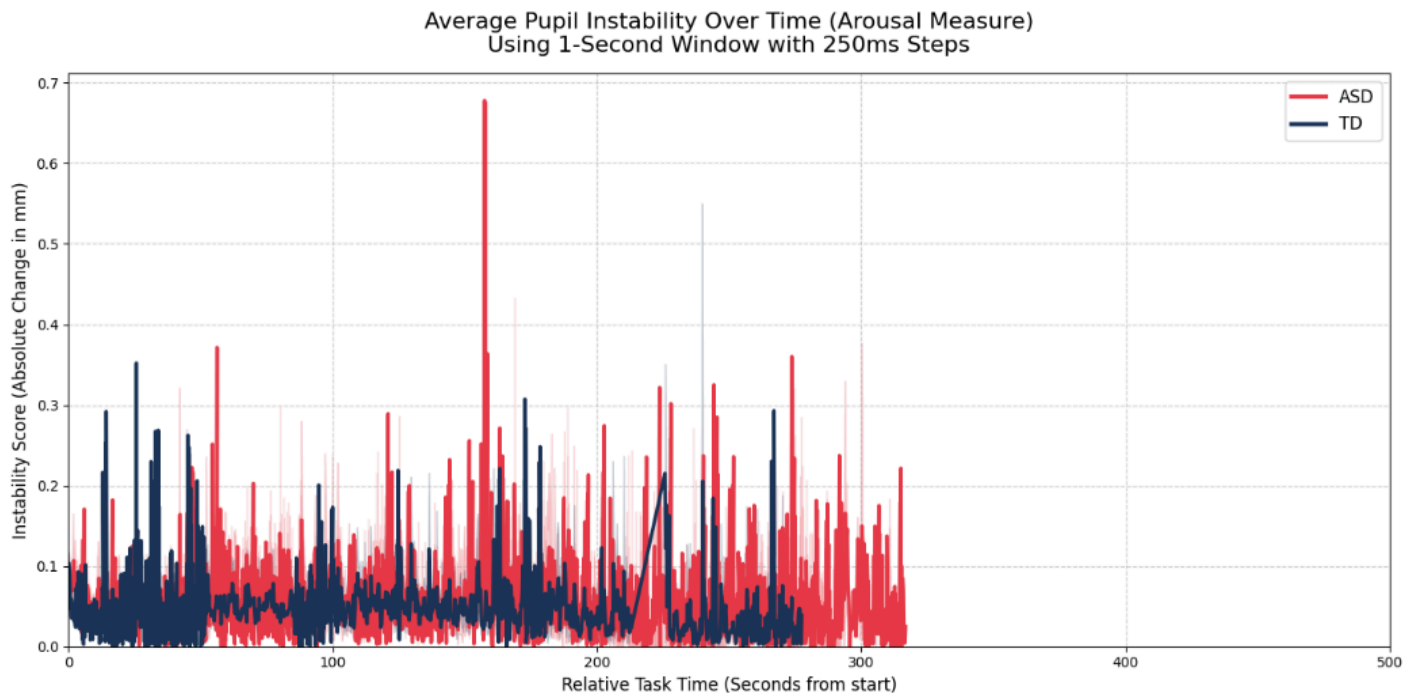


Figure 8. Average Pupil Instability Over Time by Group.

8. Challenges, Decisions, and Solutions

The main challenges and our solutions are presented in Table 7:

Challenge	Impact	Solution / Decision
Noisy and incomplete eye-tracking data	Eye-tracking data contains blinks, missing values, and inconsistent samples.	Rows with missing required values were removed, blink samples were filtered, and windows/blocks with insufficient valid samples were ignored.
A small number of participants	After preprocessing, the model was trained on 57 valid participants, which limits generalization.	The evaluation used stratified 5-fold cross-validation and results are described as prototype-level evidence rather than clinical validation.
Feature interpretability	High accuracy alone is not enough for clinical/research interpretation.	The system produces feature-pattern summaries and ablation results to explain which behavioral signals contribute.
Balancing feature groups	Combining all features may not always improve each metric, especially ASD recall.	Ablation testing was used to compare feature groups separately and avoid relying only on one final accuracy value.

9. Conclusions, Lessons Learned, and Success Metrics

9.1 Conclusions from the results

The primary contribution of this work is the development of an interpretable eye-tracking feature-extraction and analysis framework that translates raw gaze recordings into behaviorally meaningful descriptors suitable for machine-learning analysis. The results support the main project objective: measurable eye-tracking features can reveal differences between ASD and TD participants and support classification above chance. The best model achieved 74.09% cross-validation accuracy using the combined feature set. These results are consistent with the broader literature indicating that eye-tracking features can contribute to ASD/non-ASD differentiation, though further validation is still required before clinical use [Setien-Ramos et al., 2022; Eretova et al., 2024].

The results also show that no single feature family fully explains the difference between the groups. Pupil dynamics achieved the highest ASD recall, while combined features achieved the best overall accuracy. This supports the decision to combine behavioral gaze features with physiological pupil features.

The most important interpretive finding is that ASD participants showed lower fixation duration and total AOI time, suggesting less prolonged focused engagement with AOIs. At the same time, they showed higher gaze dispersion, longer saccade lengths, and more AOI changes, suggesting a broader or less concentrated scanning pattern.

9.2 Lessons learned

The lessons learned are:

- Preprocessing decisions strongly affect research conclusions, especially in physiological measures such as pupil diameter.
- Ablation testing is necessary because it shows which feature groups contribute to classification and which groups may improve only specific metrics.
- Interpretable features are valuable because they allow the results to be discussed in behavioral terms, not only as machine-learning scores.
- A small dataset requires careful wording: the project demonstrates feasibility but does not provide clinical validation.

9.3 Success metrics assessment

Metric	Assessment	Explanation
Build a working research project	Achieved	The Python code processes CSV files, extracts features, trains models, and generates outputs.
Extract meaningful eye-tracking features	Achieved	14 participant-level features were extracted across gaze behavior and pupil dynamics.
Compare ASD and TD groups	Achieved	The output includes group averages and plain-English feature-pattern summaries.
Evaluate classification performance	Achieved	5-fold cross-validation classification reports and confusion matrices were generated.

Support interpretability	Achieved	Ablation testing and feature-pattern analysis explain the contribution of feature groups.
Create a clinically deployable system	Not in current scope	The current deliverable is a research prototype, not a validated diagnostic product.

10. Software System Deliverable

- final_project_eye_tracking.py - main Python project exported from the development notebook.
- Input CSV folder - Tobii Pro Lab eye-tracking exports.
- Generated graphs - instability curves, classification reports, confusion matrices, and feature summaries.
- Documentation - this project book, user guide, maintenance guide, and poster.

Appendix A. User Guide

A.1 Input preparation

1. Open the project in Google Colab.
2. Place all Tobii-exported CSV files in one Google Drive folder.
3. Verify that the CSV files include the required columns: Participant, RecordingTime [ms], Category Right, Point of Regard Right X [px], Point of Regard Right Y [px], and Pupil Diameter Right [mm].
4. If AOI analysis is needed, verify that AOI Name Right exists in the CSV files.
5. Update the path variable in the code so it points to the CSV folder.

A.2 Running the system

1. Run the import and configuration section.
2. Mount Google Drive if the files are stored in Drive.
3. Run the pupil-instability section if the arousal graph is required.
4. Run the feature-extraction and classification section.
5. Review the printed dataset size and confirm that participants were loaded correctly.
6. Review the classification reports, confusion matrices, and feature-pattern summaries.

A.3 Outputs to review

- Dataset size and number of valid participants.
- Ablation-study accuracy values for each feature group.
- Classification report for TD and ASD precision, recall, and F1-score.
- Confusion matrices for each feature group.
- Feature-pattern summary comparing ASD and TD averages.

A.4 Interpretation notes

Accuracy measures the overall percentage of correct predictions. Recall for ASD measures the proportion of ASD participants correctly identified. Precision for ASD measures how many participants predicted as ASD were actually ASD. F1-score balances precision and recall. The confusion matrix should be used together with the classification report because it shows the actual distribution of correct and incorrect predictions.

Appendix B. Maintenance Guide

B.1 Operating environment

- Recommended environment: Google Colab.
- Required libraries: pandas, NumPy, matplotlib, seaborn, scikit-learn.
- Input format: CSV files exported from Tobii Pro Lab.

B.2 Installation and setup

In Colab, the required scientific Python libraries are usually preinstalled. In a local Python environment, install them with:

```
pip install pandas numpy matplotlib seaborn scikit-learn
```

B.3 Configuration points for maintainers

- path - location of the Tobii CSV files.
- pupil_col - name of the pupil-diameter column.
- aoi_col - name of the AOI column if available.
- Participant ID mapping - current implementation maps IDs 1-29 to ASD and 30-59 to TD.
- Feature groups - update feature indexing if new features are added.
- Model settings - Random Forest grid-search parameters can be expanded for future experiments.

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